No of neutrons in boron = 11 – 5 = 6

No of neutrons in carbon = 12 – 6 = 6

The above pair of elements Boron and Carbon has the same number of neutrons but different number of protons and hence different atomic numbers. Atoms of different elements with different atomic numbers and different mass numbers, but with same number of neutrons are called isotones

Activity 5

Draw the model of the following pairs of isotones:

(i)Fluorine & Neon (ii) Sodium & Magnesium (iii) Aluminum and Silicon

11.7 Laws of Chemical combination

In the seventeenth century, scientists had been trying to find out methods for converting one substance into another. During their studies of chemical changes, they made certain generalisations. These generalisations are known as laws of chemical combination. These are :

1. Law of conservation of mass
2. Law of constant proportions
3. Law of multiple proportions
4. Law of reciprocal proportions
5. Gay Lussac's law of gaseous volumes

Out of these five laws you have already learnt the first two laws in class VIII. Let us see the next three laws in detail in this chapter.

11.7.1 Law of multiple proportions

This law was proposed by **John Dalton in 1804**.

It states that, **"When two elements A and B combine together to form more than one compound, then different masses of A which separately combine with a fixed mass of B are in simple ratio".**

To illustrate the law let us consider the following example.

Carbon combines with oxygen to form two different oxides, carbon monoxide(CO) and carbon dioxide (CO_2). The ratio of masses of oxygen in CO and CO_2 for fixed mass of carbon is 1: 2.

	Mass of carbon (g)	Mass of oxygen (g)	Ratio of O in CO to O in CO_2
CO	12	16	1:2
CO_2	12	32	

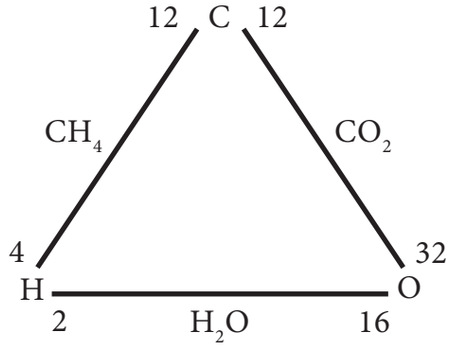
Let us take one more example, Sulphur combines with oxygen to form sulphur dioxide and sulphur trioxide. The ratio of masses of oxygen in SO_2 and SO_3 for fixed mass of Sulphur is 2:3.

11.7.2 Law of Reciprocal Proportions

The **law of reciprocal proportions** was proposed by Jeremias Ritcher in 1792.

It states that, "**If two different elements combine separately with the same weight of a third element, the ratio of the masses in which they do so are either same or a simple multiple of the mass ratio in which they combine among themselves**."

Consider the three elements hydrogen, oxygen and water as shown below:



Here, hydrogen and oxygen combine separately with the same weight of carbon to form methane (CH_4) and carbon dioxide (CO_2)

IX_SCI_EM_Unit-11_CHE.indd 133 30/11/2022 19:45:32